



Gianluca Rossi

Date of birth: 15/03/2000 | **Nationality:** Italian | **Phone number:** (+39) 3457905168 (Mobile) |
Email address: gianluca.rossi2000@gmail.com | **LinkedIn:** www.linkedin.com/in/gianlucarossi00 |
Address: 4 avenue Salvador Allende , 69100, Villeurbanne, France (Work)

WORK EXPERIENCE

PHD STUDENT – LIRIS LABORATORY AT LYON 1 UNIVERSITY – 01/10/2024 – VILLEURBANNE, FRANCE

The focus of my PhD is the combination of graph data and time series data to discover new insights and unlock analytical workflows that were previously impossible when these data types were treated in isolation.

More specifically, my work consists of:

- Defining a data model that integrates both graph data and time series data.
- Developing efficient storage mechanisms to minimize memory overhead and enable fast data retrieval.
- Designing algorithms that leverage both data types to create new analytical workflows.

UNIVERSITY TEACHING ASSISTANT – LYON 1 UNIVERSITY – 13/10/2025 – 03/03/2026 – VILLEURBANNE, FRANCE

During the second year of my PhD, I taught several courses in both English and French: Bases de données avancées (INF3048L), Big Graph Processing Systems (INF2497M), Data Processing and Analytics (INF2492M), Distributed Artificial Intelligence and Multi-Agent Systems (INF2499M), and Research Internship Training and Soft Skills 1 (INF2501M). This teaching component accounts for 66 hours of Formation à l'Insertion Professionnelle (FIP).

RESEARCH ENGINEER – LABORATOIRE D'INFORMATIQUE EN IMAGE ET SYSTÈMES D'INFORMATION – 01/03/2024 – 21/06/2024 – VILLEURBANNE, FRANCE

- Development of a new data model to combine the Property Graph Model with the Time Series model.
- Insights on Graph Sparsification techniques.

EMBEDDED SYSTEMS SOFTWARE DEVELOPER – MÈTIS VELA UNIPD – 16/10/2021 – 06/2024 – PADOVA, ITALY

I'm working on sailing boat team of University of Padova.

I am in the electronics and IT sector.

My tasks are the develop Real Time Operating System by using C++ programming language, to control the micro-controller and the sensors attached to the boat.

I developed a sensor for measuring shroud forces of the mast. This sensor is used to show to the crew, the forces applied to this shrouds.

VISITOR SERVICES REPRESENTATIVE – COMUNE DI THIENE – 10/2023 – 12/2023 – THIENE, ITALY

- Checking tickets of the customers
- Helping the them to find their assigned place

VISITOR SERVICES REPRESENTATIVE – COMUNE DI THIENE – 10/2022 – 03/2023 – THIENE, ITALY

- Checking tickets of the customers
- Helping the them to find their assigned place

VISITOR SERVICES REPRESENTATIVE – COMUNE DI THIENE – 10/2021 – 04/2022 – THIENE, ITALY

- Checking tickets of the customers
- Helping the them to find their assigned place

During my internship, I developed a web application, that allows the employees of Comune di Thiene, to search work phone numbers of other colleagues.
I did the client-side part using raw HTML, CSS and JAVASCRIPT.
For the server-side, I used php for the login process of the users and for the management of the database, that holds all the record composed by:
employee name, phone number, office.

● EDUCATION AND TRAINING

28/09/2022 – 03/09/2024 Padova, Italy

MASTER IN COMPUTER ENGINEER

Address Via VIII Febbraio , 35122, Padova, Italy | **Website** <https://www.unipd.it/> | **Final grade** 110/110 |

Thesis Unifying subgraph matching and subsequence joins in hybrid graphs

05/01/2024 – 21/06/2024 Villeurbanne, France

ERASMUS STUDENT Université Claude Bernard Lyon 1

Website <https://www.univ-lyon1.fr/>

01/10/2019 – 21/10/2022 Padova, Italy

BACHELOR IN COMPUTER ENGINEER University of Padova

Address Via VIII Febbraio , 2, 35122, Padova, Italy | **Website** <https://www.unipd.it/> | **Field of study** Computer Engineering |

Final grade 96/110 | **Thesis** Acquisizione dati da celle di carico per telemetria di barche a vela del Progetto Mètis Vela

15/09/2014 – 26/06/2019 Thiene, Italy

IT SYSTEM MANAGER AND IT PROGRAMMER ITT G. Chilesotti

Address Via Tigli, 10, 36016, Thiene, Italy | **Website** <https://www.chilesotti.edu.it/> | **Final grade** 70/100

● PUBLICATIONS

2026

Unifying Graph Traversals and Time Series Joins in Hybrid Graphs

In this work, we unify graph traversals and time series analysis, introducing the first declarative subgraph matching approach for hybrid graphs. By combining subsequence joins with Allen's interval algebra, we enable graph traversals that jointly capture structural and temporal patterns.

Authors: Gianluca Rossi, Angela Bonifati and Riccardo Tommasini | **Journal Name:** IEEE ICDE 2026

2025

[TD-Join: Leveraging Temporal Dependencies in Time Series Joins](#)

Current data mining algorithms for time series joins focus only on finding the best subsequences, completely neglecting the temporal dependencies present in time series. We propose TD-Join, a novel tool for temporal reasoning on time series. TD-Join enhances time series subsequence joins by integrating Allen's Algebra, a widely adopted framework for temporal reasoning. Our approach enables users to efficiently identify, query, and interpret temporal relationships over similar time series subsequences. Write here the description...

Companion of the 2025 International Conference on Management of Data (SIGMOD-Companion '25), June 22--27, 2025, Berlin, Germany

Authors: Gianluca Rossi, Riccardo Tommasini and Angela Bonifati | **Journal Name:** ACM Sigmod 2025 | **Volume, Issue and Pages:** Pages 223 - 226 | **Publisher:** ACM

Link <https://gianlucarossi15.github.io/TD-Join/>

2025

[Reproducibility Report for ACM SIGMOD 2025 Paper: "SuSe: Summary Selection for Regular Expression Subsequence Aggregation over Streams"](#)

The paper was submitted for the LongEval-Retrieval task of the CLEF 2023 conference. The aim of the LongEval-Retrieval task is to measure how much information retrieval systems are resilient when the training data used to train the retrieval model is very distant (in terms of time) with respect to the test set.

● CONFERENCES AND SEMINARS

04/05/2026 – 08/05/2026 Montreal, Canada

42nd IEEE International Conference on Data Engineering (ICDE)

I attended the ICDE 2026 conference to present my research paper *Unifying Graph Traversals and Time Series Joins in Hybrid Graphs*. The experience provided a valuable opportunity to engage with researchers in my field, exchange ideas, and receive insightful feedback that will help further strengthen my work.

07/07/2025 – 11/07/2025 Nicosia, Cyprus

EDBT 2025 Summer School

EDBT summer school about artificial intelligence and data management, with a special focus on Large Language Models, AI agents and Vector Databases.

Link <https://dmai.cs.ucy.ac.cy/index.html>

22/06/2025 – 27/06/2025 Berlin

ACM SIGMOD/PODS International Conference on Management of Data

I attended Sigmod/Pods 2025 conference on data management, held in Berlin, where I had the opportunity to present my short paper **TD-Join: Leveraging Temporal Dependencies in Time Series Joins**. I also had the pleasure to meet and discuss with top researcher to discuss about databases.

Link <https://2025.sigmod.org/index.shtml>

18/09/2023 – 21/09/2023 Thessaloniki

24th CLEF Conference and Labs of the Evaluation Forum

I had the opportunity to present remotely my paper for the LongEval-Retrieval task of the LongEval laboratory.

Link <https://clef2023.clef-initiative.eu/index.php>

● LANGUAGE SKILLS

Mother tongue(s): **ITALIAN**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	B2	B2	B2	B2
FRENCH	B2	B1	A2	A2	A2
SPANISH	B1	B2	B1	B1	B1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

● SKILLS

Social Media | Microsoft Office | Zoom | Google Drive | LaTeX (basic) | Skype

● DRIVING LICENCE

Driving Licence: AM

Driving Licence: A1

Driving Licence: B

● **HOBBIES AND INTERESTS**

Hobbies

I am addicted for skiing, racing cars and djing. I am also a good swimmer.

● **VOLUNTEERING**

04/2024 – 05/2024 Lyon

Food distribution volunteer Lyon Heat

Food distribution for needy students

● **COMMUNICATION AND INTERPERSONAL SKILLS**

Interpersonal skills

I am a very determined guy, always pursuing my goals.

Everytime I do my best to accomplish the tasks that I encounter. I love working in team, sharing different ideas.